- **Subtitle:** Task №3

- **Team name:** Mohajer UAV

Members:

- Sobhan Memarzadeh
- Hossein Shirani
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- Mehdi Aghaziyarati

Country : IRAN





What is Task 3? (Swarming Drones)



ТЕХНИК 6S from Skyris Company



Technic-Show For Swarming Drones



Technic Drone Show

Selected drones | Server | Table

copter ID	version	configuration	animation ID	battery	FCU status	sensors	mode	checks	current x y z yaw frame_id	start x y z ya

Start after · s

Preflight check

Start animation

Pause

Land selected

Land ALL

Emergency land

Disarm ALL

Disarm selected

Test leds

Takeoff

Z = \/· m

Flip

Reboot FCU

Calibrate gyro

Calibrate level

But we are looking for the best



TECHNIC | Drone Show Command Center

Selected drones Server Table

TECHNIC
DRONE SHOW / COMMAND CENTER

Visual motion

چهارشنبه ر ۱۱ نوامبر | ۰۹:۰۴:۴۱

TECHNIC / FLIGHT OPERATIONS

Precision in every flight.

Plan with clarity. Command with confidence.

DECORATIVE ARTWORK / NOT LIVE TELEMETRY

FLEET RECORDS
0
Rows in the real fleet model

SELECTED DRONES
0
Command checkbox selection

SELECTED / READY
0 / 0
Existing preflight checks

FLEET / LIVE TABLE

Original telemetry model • original status colours

Select all Clear Columns

NO FLEET RECORDS
Waiting for drone data — no demo rows are added.

Selection uses the original checkboxes, not highlighted table cells.

COMMAND DECK

01 / SHOW SEQUENCE
Uses the original preflight and start checks.
Start after • s
Preflight check
Start animation
Pause

02 / FLIGHT CONTROL
Scope: selected drones
Takeoff
Z = 1/• m
Land selected
Disarm selected
Test leds
Flip

03 / SAFETY COMMANDS
Land ALL / Disarm ALL: broadcast.
Emergency land: selected drones only.
Land ALL
Disarm ALL
Emergency land

TECHNIC / NATIVE PyQt5 INTERFACE

Command scope & safety checks retained



But I have some bad news.

Task №3

But; I also have some good news.





There is no time, and we must be quick.

TECHNIC — Flight Operations Workspace (TCP)

T /
TECHNIC
FLIGHT SYSTEMS

WORKSPACE
- Overview
- Fleet nodes
- Live stream

Operations overview

COMMAND CENTER / TCP BUS WORKSPACE

YOLO READY | HARDWARE CONTROL | ۱۲ نوامبر ۲۰۲۶ / ۱۱:۴۴:۴۵

TECHNIC / FLIGHT OPERATIONS

Precision.
In formation.

One leader. A connected fleet. A single workspace.

• 2 LINKS ONLINE / 2 REGISTERED DEVICES



QUAD / X4 · CONCEPT VISUAL

REGISTERED	ONLINE	SELECTED	RX / SECOND
2	2	2	40
Devices in workspace	Active WebSocket links	Current command scope	Incoming messages

FLEET TOPOLOGY / CONNECTION STATUS

SCHEMATIC · NOT A FLIGHT MAP



LIVE MAP / 7.5 × 4.5 m

Marker IDs

+ Add object

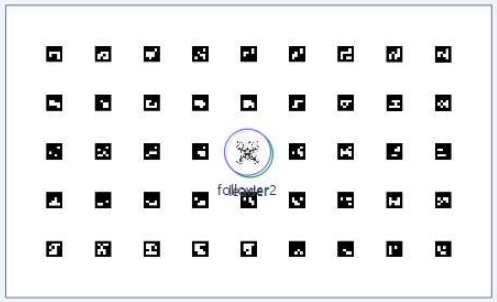
AIRCRAFT 02

leader
+0.02, -0.04 m

follower2
-0.02, -0.01 m

OBJECTS 00

+X right | +Y up | metres



COMMAND DECK

JSONL CONTROL

01 / LEADER COMMANDS

Dedicated coordinator control

Waiting for registered leader control link

Scan area

Return to nest

Immediate land

02 / SHOW SEQUENCE

Leader-camera damage alerts (color heuristic, not a diagnosis). Hardware inspection means a sequential flight visit, not an automatic diagnosis. Native pump/spray is not defined and is disabled. Verify shared map/controller origin before flight.

Follower 1 + Follower 2 (alternate)

Spray pending trees · sequential

All drones · inspect sequentially

Waiting for the leader to detect infected trees...

02 / SHOW SEQUENCE

Leader-camera damage alerts (color heuristic, not a diagnosis). Hardware inspection means a sequential flight visit, not an automatic diagnosis. Native pump/spray is not defined and is disabled. Verify shared map/controller origin before flight.

Follower 1 + Follower 2 (alternate)

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All drones · inspect sequentially

Waiting for the leader to detect infected trees...

03 / FLIGHT CONTROL

Take Off ALL · corners + leader scan

Selected controls use native TCP. Disarm and Flip are unsupported.

Leader Drone



WORKSPACE

Overview

Fleet nodes

Live stream

01 / LEADER NODE

Single coordinator · one device maximum

HOST 192.168.0.186 POSE/ROS PORT ۹۰۹۰ + Add leader

SELECT | DEVICE / LIVE TELEMETRY | LIVE CAMERAS

LEADER / COORDINATOR

192.168.0.186:9090

• ONLINE / WEBSOCKET

Battery 24.3 V (78%)

Altitude 0.04 m

118 available topics

Inspect Remove

LIVE · YOLO

LIVE

Stop cameras Configure

NO MAP FIX: 0 markers; need >=4 non-collinear

01 / LEADER COMMANDS

Dedicated coordinator control

Waiting for registered leader control link

Scan area

Return to nest

Immediate land

02 / SHOW SEQUENCE

Leader-camera damage alerts (color heuristic, not a diagnosis). Hardware inspection means a sequential flight visit, not an automatic diagnosis. Native pump/spray is not defined and is disabled. Verify shared map/controller origin before flight.

Follower 1 + Follower 2 (alternate)

Spray pending trees · sequential

All drones · inspect sequentially

Waiting for the leader to detect infected trees...

02 / CHILD FLEET

Multiple devices · select the targets for each demo action

03 / FLIGHT CONTROL

Take Off ALL · corners + leader scan

Selected controls use native TCP. Disarm and Flip are unsupported.

LIVE STREAM / WEBSOCKET CONSOLE

Pause display Clear • LISTENING / 2 LINKS

```
[11:37:29.065] [192.168.0.86:9090] {"op": "publish", "topic": "/mavros/local_position/pose", "msg": {"header": {"seq": 5647, "stamp": {"secs": 1794470848, "nsecs": 901641024}, "frame_id": "map"}, "pose": {"position": {"x": -0.042644161731004715, "y": -0.024195805191993713,
[11:37:29.065] [192.168.0.86:9090] {"op": "publish", "topic": "/mavros/altitude", "msg": {"header": {"seq": 1882, "stamp": {"secs": 1794470848, "nsecs": 880399536}, "frame_id": "map"}, "monotonic": null, "amsl": null, "local": 0.019177282229065895, "relative": 0.0191772822
```

PREFERENCES

Visual motion

NATIVE CONTROL
.240:4000
.186:5000
.86:3000
Verify state / DONE.

WORKSPACE / 02

• LOCAL WORKSPACE

YOLO ready · classes: tree / Native JSONL control

Follower drone



WORKSPACE

Overview

Fleet nodes

Live stream

HOST 192.168.0.86

POSE/ROS PORT ۹۰۹۰

+ Add child

SELECT | DEVICE / LIVE TELEMETRY | LIVE CAMERAS

CHILD / FLEET MEMBER

192.168.0.86:9090

• ONLINE / WEBSOCKET

Battery 24.2 V (77%)

Altitude -0.01 m

118 available topics

Inspect | Remove

LIVE · YOLO

CONNECTING

NO CAMERA FRAMES

Start camera to connect

Stop cameras | Configure

S1 LIVE (YOLO) · S2 ...

Select all devices | Clear selection

01 / LEADER COMMANDS

Dedicated coordinator control

Waiting for registered leader control link

Scan area

Return to nest

Immediate land

02 / SHOW SEQUENCE

Leader-camera damage alerts (color heuristic, not a diagnosis). Hardware inspection means a sequential flight visit, not an automatic diagnosis. Native pump/spray is not defined and is disabled. Verify shared map/controller origin before flight.

Follower 1 + Follower 2 (alternate)

Spray pending trees · sequential

All drones · inspect sequentially

Waiting for the leader to detect infected trees...

NETWORK ACTIVITY / LIVE RX

12,392 received · actual WebSocket messages, not flight telemetry

03 / FLIGHT CONTROL

Take Off ALL · corners + leader scan

Selected controls use native TCP. Disarm and Flip are unsupported.

LIVE STREAM / WEBSOCKET CONSOLE

Pause display | Clear

• LISTENING / 2 LINKS

```
[11:38:07.020] [192.168.0.86:9090] {"op": "publish", "topic": "/mavros/local_position/pose", "msg": {"header": {"seq": 6785, "stamp": {"secs": 1794470886, "nsecs": 837275968}, "frame_id": "map"}, "pose": {"position": {"x": -0.05087044625616074, "y": -0.02505413442850113,
[11:38:07.020] [192.168.0.86:9090] {"op": "publish", "topic": "/mavros/altitude", "msg": {"header": {"seq": 2262, "stamp": {"secs": 1794470886, "nsecs": 879805968}, "frame_id": "map"}, "monotonic": null, "amsl": null, "local": -0.01282454188913107, "relative": -0.012824541
```

PREFERENCES

Visual motion

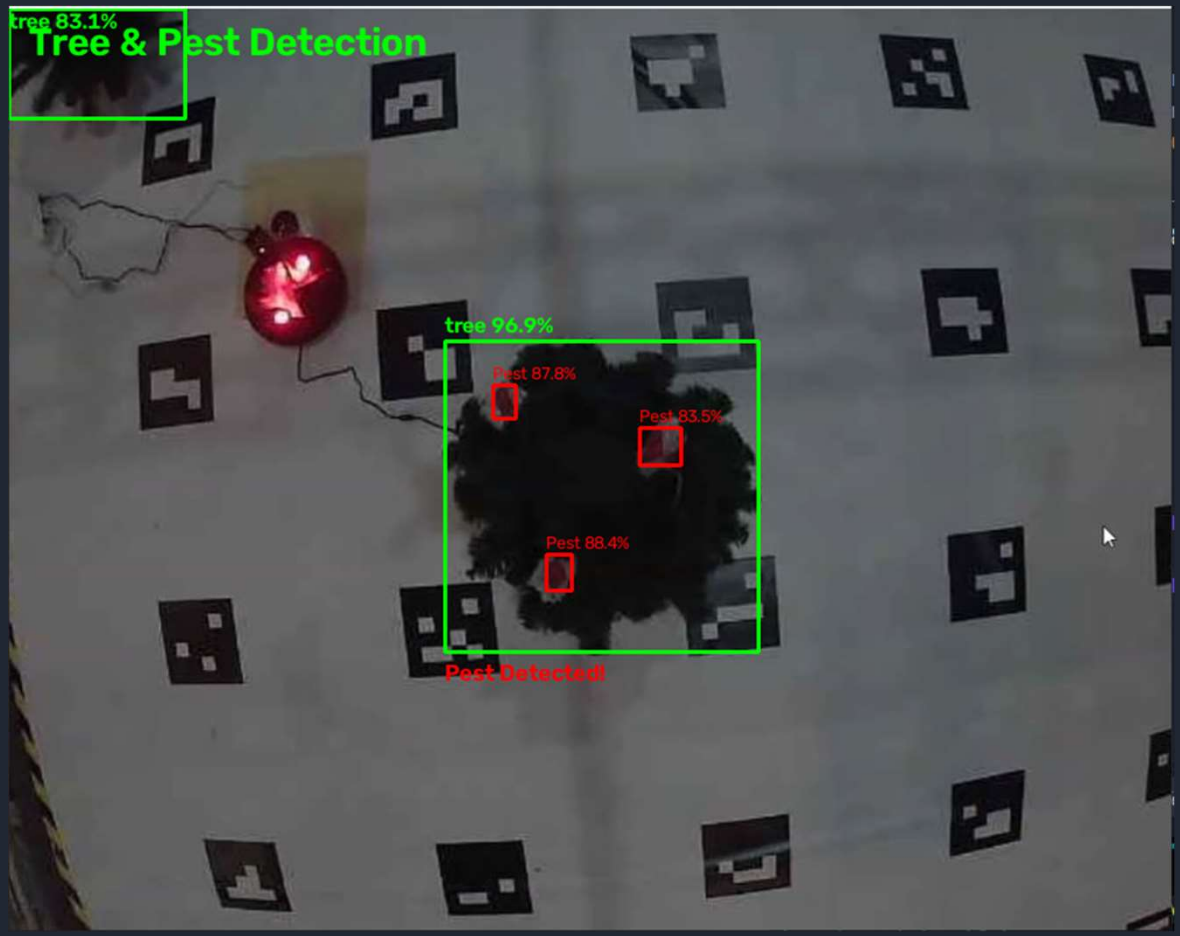
NATIVE CONTROL
.240:4000
.186:5000
.86:3000
Verify state / DONE.

WORKSPACE / 02

• LOCAL WORKSPACE

YOLO ready · classes: tree / Native JSONL control

Drone Detection



Drone Detection



Drone Detection



T /
TECHNIC
FLIGHT SYSTEMS

WORKSPACE
Overview
Fleet nodes
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Operations overview
COMMAND CENTER / TCP BUS WORKSPACE

YOLO READY | HARDWARE CONTROL | ۱۲ نوامبر ۲۰۲۶ / ۱۱:۴۷:۳۶

Selected devices | Ping all | Read state | Clear

01 / LEADER NODE
Single coordinator · one device maximum

HOST 192.168.0.186 | POSE/ROS PORT ۹۰۹۰ | + Add leader

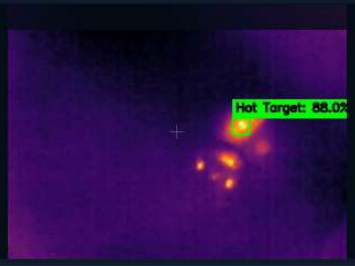
SELECT | DEVICE / LIVE TELEMETRY | LIVE CAMERAS

LEADER / COORDINATOR
192.168.0.186:9090
• ONLINE / WEBSOCKET
Battery 24.0 V (72%)
Altitude 1.52 m
118 available topics
Inspect | Remove

LIVE · YOLO



LIVE



Stop cameras | Configure

NO MAP FIX: 5 markers; need >=4 non-collinear

02 / CHILD FLEET
Multiple devices · select the targets for each demo action

COMMAND DECK
JSONL CONTROL

01 / LEADER COMMANDS
Dedicated coordinator control
Waiting for registered leader control link
Scan area
Return to nest
Immediate land

02 / SHOW SEQUENCE
Leader-camera damage alerts (color heuristic, not a diagnosis). Hardware inspection means a sequential flight visit, not an automatic diagnosis. Native pump/spray is not defined and is disabled. Verify shared map/controller origin before flight.

Follower 1 + Follower 2 (alternate)
Spray pending trees · sequential
All drones · inspect sequentially
Waiting for the leader to detect infected trees...

03 / FLIGHT CONTROL
Take Off ALL · corners + leader scan
Selected controls use native TCP. Disarm and Flip are unsupported.

Drone Map



T /
TECHNIC
FLIGHT SYSTEMS

WORKSPACE
Overview
Fleet nodes
Live stream

Operations overview
COMMAND CENTER / TCP BUS WORKSPACE

YOLO READY | HARDWARE CONTROL | ۱۲ نوامبر ۲۰۲۶ / ۱۱:۴۰:۲۵

LIVE MAP / 7.5 × 4.5 m
Marker IDs
+ Add object
+X right | +Y up | metres

AIRCRAFT 01
leader
-1.81, -0.54 m

OBJECTS 07
Tree · cam5c2eec8cd91
-1.25, +0.51 m
Fire · cam5c2eec8cd91
-2.49, +1.03 m
Tree · cam5c2eec8cd91
-0.04, +1.16 m

leader

Origin · marker 122 / Metres / X → Y ↑

COMMAND DECK
JSONL CONTROL

01 / LEADER COMMANDS
Dedicated coordinator control
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Leader-camera damage alerts (color heuristic, not a diagnosis). Hardware inspection means a sequential flight visit, not an automatic diagnosis. Native pump/spray is not defined and is disabled. Verify shared map/controller origin before flight.
Follower 1 + Follower 2 (alternate)
Spray pending trees · sequential
All drones · inspect sequentially
Waiting for the leader to detect infected trees...

03 / FLIGHT CONTROL
Take Off ALL · corners + leader scan
Selected controls use native TCP. Disarm and Flip are unsupported.

CONTROL SOCKET / JSON LINES
× CONTROL 0/1 connected
Native control: .240 → 4000 / .186 → 5000 / .86 → 3000. QUEUED is not flight completion; check TX, state and DONE. Map poses still come from each drone's HTTP/ROS source.

```
[11:44:10.749] i control       LINK [leader 192.168.0.240:4000] disconnected: timed out
[11:44:11.746] i control       LINK [leader 192.168.0.240:4000] connecting
[11:44:13.814] i control       LINK [leader 192.168.0.240:4000] disconnected: timed out
[11:44:14.798] i control       LINK [leader 192.168.0.240:4000] connecting
[11:44:16.077] i control       LINK [follower1 192.168.0.186:5000] connecting
[11:44:16.766] i control       LINK [leader 192.168.0.240:4000] disconnected: timed out
[11:44:17.756] i control       LINK [leader 192.168.0.240:4000] connecting
[11:44:18.087] i control       LINK [follower1 192.168.0.186:5000] disconnected: timed out
[11:44:19.182] i control       LINK [follower1 192.168.0.186:5000] connecting
[11:44:19.727] i control       LINK [leader 192.168.0.240:4000] disconnected: timed out
[11:44:20.814] i control       LINK [leader 192.168.0.240:4000] connecting
[11:44:21.139] i control       LINK [follower1 192.168.0.186:5000] disconnected: timed out
[11:44:22.129] i control       LINK [follower1 192.168.0.186:5000] connecting
```

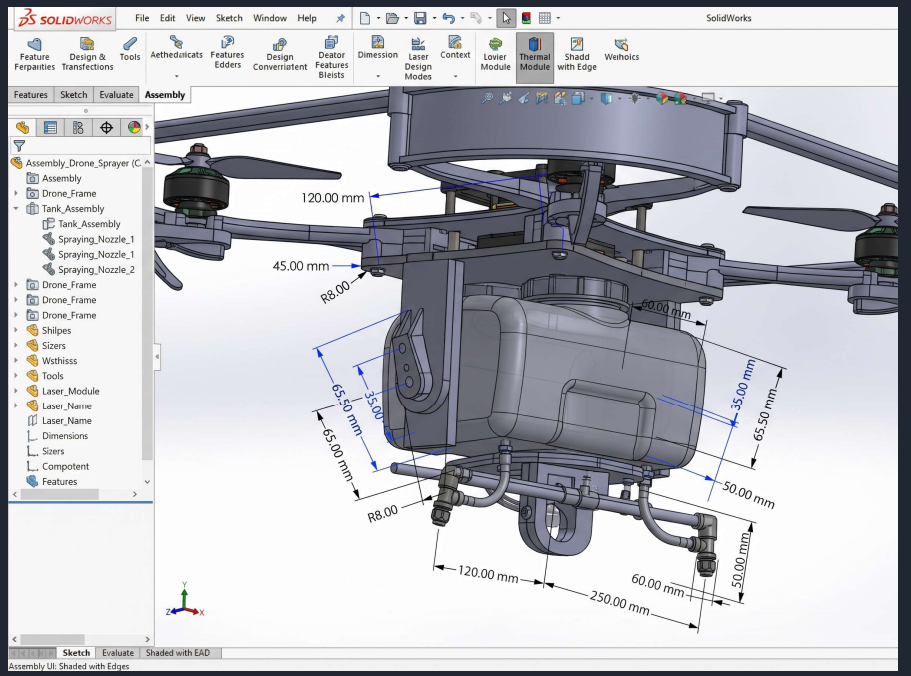
Final Task №3



We always think about the future as well.

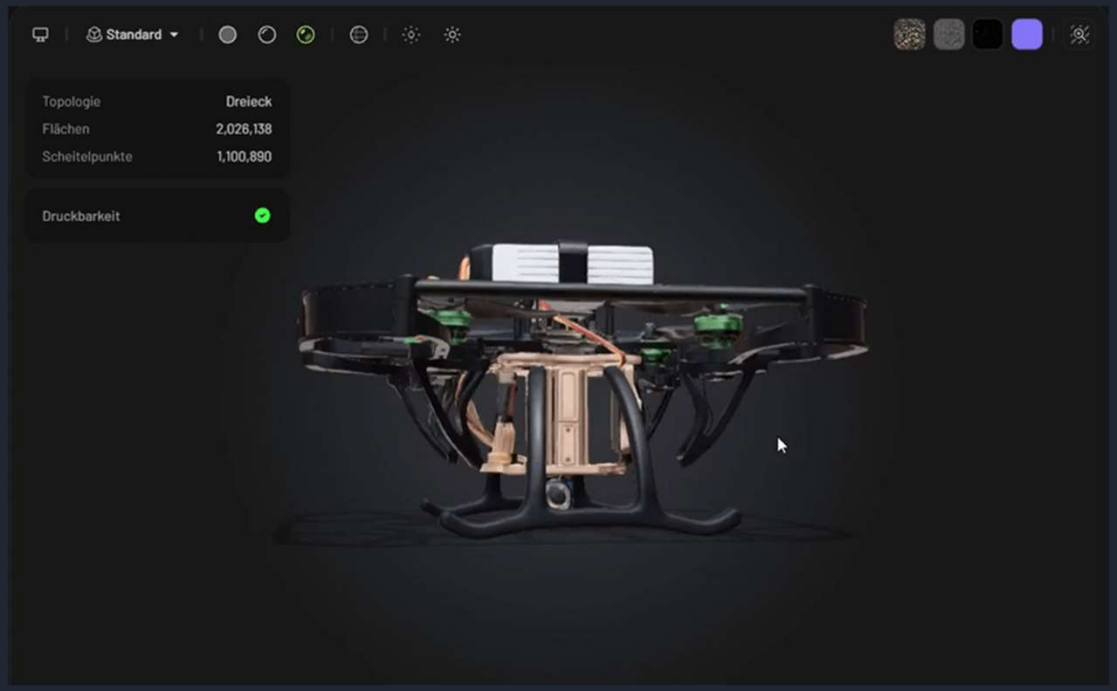


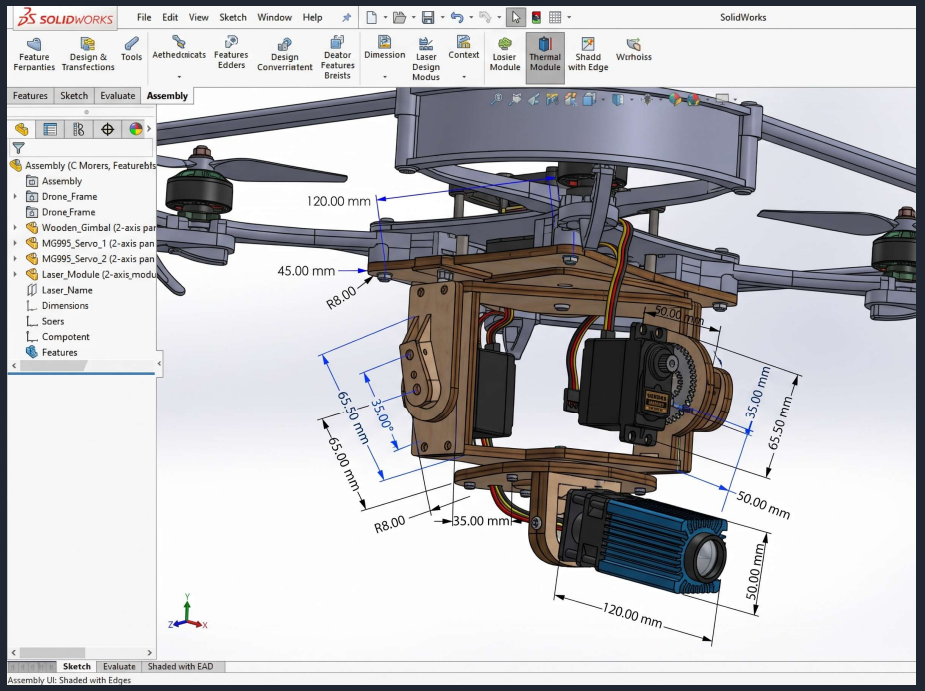




We always think about the future as well.



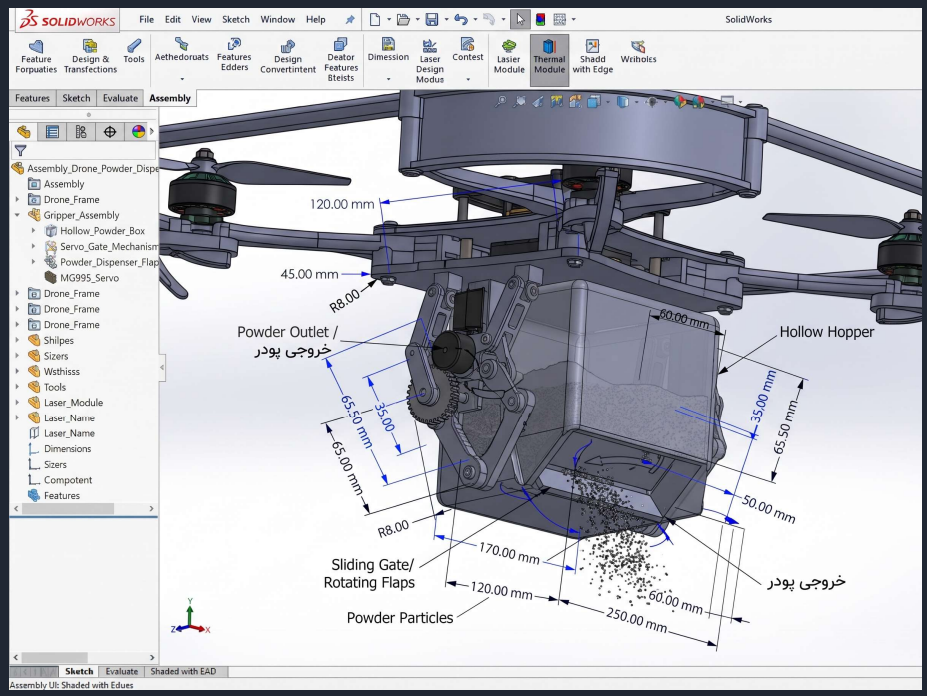




We always think about the future as well.







Thank you for your attention



